

Personal Knowledge Base (Second Brain)

Project Description

Personal Knowledge Base (Second Brain) is a backend-based knowledge management system developed using Node.js, Express.js, and MongoDB. The system allows users to store, organize, search, and manage personal knowledge such as notes, ideas, learning resources, article summaries, project information, and research materials in a centralized repository.

The project is implemented as a REST API application without any frontend interface. All operations are performed through API endpoints and tested using Postman or Swagger. MongoDB is used as the database to store and manage knowledge documents efficiently.

Objective

The primary objective of this project is to provide a structured platform for capturing and retrieving information while demonstrating MongoDB concepts such as document storage, CRUD operations, indexing, aggregation, and relationship mapping.

Working of the System

A user first registers and authenticates through REST APIs. After successful authentication, the user can create notes containing a title, content, category, tags, creation date, and ownership information. These notes are stored as documents in MongoDB.

Users can organize notes into categories such as Programming, Product Management, Research, Career Planning, and College Notes. Multiple tags can also be assigned to each note to improve classification and retrieval.

The system provides REST APIs for Create, Read, Update, and Delete (CRUD) operations on notes, categories, and tags. Whenever a request is received, Express.js processes the request, applies business logic, interacts with MongoDB through Mongoose, and returns a JSON response.

To improve retrieval efficiency, MongoDB text indexes are created on note titles and content. Users can search notes using keywords, categories, tags, favorite status, or creation dates. The search API queries MongoDB and returns matching documents instantly.

The system also provides analytical APIs that generate insights such as total notes created, most used tags, notes per category, recently added notes, and favorite note counts using MongoDB aggregation pipelines.

Database Relationships

The project demonstrates MongoDB relationship mapping:

- One-to-Many: One user can create multiple notes.
- Many-to-One: Multiple notes can belong to a single category.
- Many-to-Many: Notes can contain multiple tags, and a tag can be associated with multiple notes.

Technical Stack

Backend: Node.js, Express.js

Database: MongoDB

ODM: Mongoose

API Testing: Postman / Swagger

Version Control: Git & GitHub

Conclusion

The Personal Knowledge Base (Second Brain) provides a scalable and efficient solution for personal knowledge management. The project demonstrates practical implementation of REST APIs, MongoDB document modeling, indexing, aggregation, CRUD operations, and relationship mapping while creating a searchable digital repository for long-term knowledge storage and retrieval.